<Company Name>

<Project Name>
Software Requirements Specifications

Version <1.0>

[Note: The following template is provided for use with the Unified Process for EDUcation. Text enclosed in square brackets and displayed in blue italics (style=InfoBlue) is included to provide guidance to the author and should be deleted before publishing the document. A paragraph entered following this style will automatically be set to normal (style=Body Text).]

[To customize automatic fields in Microsoft Word (which display a gray background when selected), select File>Properties and replace the Title, Subject and Company fields with the appropriate information for this document. After closing the dialog, automatic fields may be updated throughout the document by selecting Edit>Select All (or Ctrl-A) and pressing F9, or simply click on the field and press F9. This must be done separately for Headers and Footers. Alt-F9 will toggle between displaying the field names and the field contents. See Word help for more information on working with fields.]

Revision History

Date	Version	Description	Author
3/7/2026	0.1	Section 3.1	Austin Haight

Table of Contents

Software Requirements Specifications

1. Introduction

*[The introduction of the **Software Requirements Specification (SRS)** provides an overview of the entire **SRS**. It includes the purpose, scope, definitions, acronyms, abbreviations, references, and overview of the **SRS**.]*

*[Note: The **SRS** captures the complete software requirements for the system, or a portion of the system. Following is a typical **SRS** outline for a project **using use-case modeling**. This artifact consists of a package containing use cases of the use-case model and applicable Supplementary Specifications and other supporting information.]*

*[Many different arrangements of an **SRS** are possible. Refer to [IEEE830-1998] for further elaboration of these explanations, as well as other options for **SRS** organization.]*

1.1 Purpose

*[Specify the purpose of this **SRS**. The **SRS** fully describes the external behavior of the application or subsystem identified. It also describes nonfunctional requirements, design constraints, and other factors necessary to provide a complete and comprehensive description of the requirements for the software.]*

1.2 Scope

*[A brief description of the software application that the **SRS** applies to, the feature or other subsystem grouping, what Use-Case model(s) it is associated with, and anything else that is affected or influenced by this document.]*

1.3 Definitions, Acronyms, and Abbreviations

*[This subsection provides the definitions of all terms, acronyms, and abbreviations required to properly interpret the **SRS**. This information may be provided by reference to the project's Glossary.]*

1.4 References

*[This subsection provides a complete list of all documents referenced elsewhere in the **SRS**. Identify each document by title, report number if applicable, date, and publishing organization. Specify the sources from which the references can be obtained. This information may be provided by reference to an appendix or to another document.]*

1.5 Overview

*[This subsection describes what the rest of the **SRS** contains and explains how the document is organized.]*

2. Overall Description

*[This section of the **SRS** describes the general factors that affect the product and its requirements. This section does not state specific requirements. Instead, it provides a background for those requirements, which are defined in detail in Section 3, and makes them easier to understand. Include such items as:*

2.1 Product perspective

2.1.1 System Interfaces

2.1.2 User Interfaces

2.1.3 Hardware Interfaces

2.1.4 Software Interfaces

2.1.5 Communication Interfaces

2.1.6 Memory Constraints

2.1.7 Operations

2.2 Product functions

2.3 User characteristics

2.4 Constraints

2.5 Assumptions and dependencies

2.6 Requirements subsets

3. Specific Requirements

*[This section of the **SRS** contains all software requirements to a level of detail sufficient to enable designers to design a system to satisfy those requirements, and testers to test that the system satisfies those requirements. When using use-case modeling, these requirements are captured in the Use Cases and the applicable supplementary specifications. If use-case modeling is not used, the outline for supplementary specifications may be inserted directly into this section, as shown below.]*

3.1 Functionality

*[This section describes the functional requirements of the system for those requirements that are expressed in the natural language style. For many applications, this may constitute the bulk of the **SRS** package and thought should be given to the organization of this section. This section is typically organized by feature, but alternative organization methods may also be appropriate; for example, organization by user or organization by subsystem. Functional requirements may include feature sets, capabilities, and security.*

Where application development tools, such as requirements tools, modeling tools, and the like, are employed to capture the functionality, this section of the document would refer to the availability of that data, indicating the location and name of the tool used to capture the data.]

3.1.1 User Interface

The program must have a clean, readable user interface. Interface must include a field to enter expressions either by labeled buttons or by keyboard input. Available somewhere should be a list of operators and their function.

3.1.2 History/Memory

The program should keep a number of past expressions from the current session, and make them available to the user. The user should be able to easily pull values from past expressions to use in the current one.

3.1.3 Token Parsing

The program must be able to turn any syntactically valid string into an ordered list of token objects, which can be evaluated mathematically. Tokens may be separated by an arbitrary number of space characters. Tokens must be evaluated in PEMDAS order. Tokens include:

-Integer: Any length of digits, must not contain a period. May have a sign prefix but otherwise is assumed positive valued. Regex: [+-]?[0-9][0-9]*

-Floating Point: Any two lengths of digits [0-9] with exactly one period. May have a sign prefix but otherwise is assumed positive valued. Regex: [+-]?([0-9][0-9]*\.[0-9]*|[0-9]*\.[0-9][0-9]*)

-Binary Addition: Two expressions separated by a ‘+’ operator. Adds the results of the two expressions. Resolves to an Integer only if both operands are Integers, otherwise Floating Point.

-Binary Subtraction: Two expressions separated by a ‘-’ operator. Subtracts the result of the second expression from the result of the first expression. Resolves to an Integer only if both operands are Integers, otherwise Floating Point.

-Binary Multiplication: Two expressions separated by a ‘*’ operator. Multiplies the results of the expressions. Resolves to an Integer only if both operands are Integers, otherwise Floating Point.

-Binary Division: Two expressions separated by a ‘/’ operator. Divides the result of the second expression from the result of the first expression. Resolves to a Floating Point.

-Binary Floor Division: Two expressions separated by a ‘//’ operator. Performs Binary Division and then truncates the result into an Integer.

-Modulus: Two expressions separated by a ‘%’ operator. For input A%B the result is equivalent to A-((A//B)*A). Resolves to an Integer only if both operands are Integers, otherwise Floating Point.

-Parentheses: Any expression enclosed by parentheses ‘(‘, ‘)’. Forces priority in evaluation.

3.2 Use-Case Specifications

[In use-case modeling, the use cases often define the majority of the functional requirements of the system, along with some non-functional requirements.]

3.3 Supplementary Requirements

[Supplementary Specifications capture other requirements, e.g., non-functional requirements and development constraints, that are not included in the use cases and non-functional requirements.]

4. Classification of Functional Requirements

[List, usually in a table, all functional requirements and order them by Type (Essential, Desirable, and Optional) or by order of appearance in the document.]

Functionality	Type
...	
...	

5. Appendices

*[When appendices are included, the **SRS** should explicitly state whether or not the appendices are to be considered part of the requirements]*